

Alison Brayan Laura Ordoñez

ROS 2 Robotics Engineer — Autonomous Mobile Robots, Navigation & Simulation

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SUMMARY

Mechatronics Engineer (UNI, Peru) with 5+ years building robotics software on ROS 1 and ROS 2 for autonomous mobile robots in logistics, agriculture and construction. Hands-on with SLAM, localization, sensor fusion, camera-based perception on Jetson Nano, URDF/TF2 and Gazebo simulation, in C++ and Python on Linux. Founder of Kalman Robotics: ROS 2 robot kit, simulations, a containerized browser-based ROS 2 lab, and 2,000+ engineering students trained across six universities. 1st place at the International Startup Competition (ISC) with an autonomous navigation robot.

TECHNICAL SKILLS

ROS / Robotics	ROS 2 (Humble, Galactic), ROS 1, rclcpp / rclpy, Nav2 , BehaviorTree.CPP , Cartographer, micro-ROS, TF2, URDF/xacro, RViz, Gazebo, colcon
Autonomy	SLAM, localization, autonomous navigation, odometry, differential & mecanum drive kinematics, trajectory tracking, robot control
Perception	Computer vision (YOLO, Intel RealSense D435i), Jetson Nano, Kalman filters, sensor fusion
Sensors	LiDAR, IMU, wheel encoders, USB/V4L2 cameras; sensor integration and coordinate frames
Programming	C++, Python, CMake, Bash, JavaScript, SQL
Platforms & Tools	Linux (Ubuntu), Docker / Docker Compose, Jenkins CI (automated ROS tests), Git, Vue.js web interfaces, MATLAB/Simulink
Embedded / HW	Jetson Nano, Raspberry Pi, STM32, ESP32, PIC, ATmega; hardware–firmware–ROS integration

PROFESSIONAL EXPERIENCE

Founder & CEO — Kalman Robotics S.A.C. 2024 – Present

Robotics education startup: ROS 2 hardware kit, simulators and online courses — kalmanrobotics.io

- Designed the ROS 2 software stack for the Kalman educational robot kit: bringup, URDF description, Gazebo simulation, custom interfaces, IMU, telemetry and teleoperation packages.
- Built Gazebo simulations of mobile robots (Kalman Nexus, TortoiseBot) used as the base for hands-on training.
- Leading **Kalman Lab**, a browser-based ROS 2 lab: one Docker container per student with VS Code, terminal and a remote desktop running Gazebo and RViz.
- Produced video courses on ROS 2, Linux, URDF and CMake with a Python automation pipeline; delivered university workshops.

Head of Robotics Development — Plural Soft Mar 2023 – Oct 2023

- Led development of a **forklift-type autonomous mobile robot** prototype for warehouse logistics.
- Developed the robot software in ROS 2 (Galactic), validated it in Gazebo, and integrated it with the robot's hardware.

Head of Robotics Development & Innovation — Ruway Tec Jan 2021 – Feb 2023

- Developed autonomous navigation for Leo Rover and MecanumBot platforms in ROS 1 and ROS 2, with Gazebo simulation.
- Led Industry 4.0 projects for mining: laser drilling-pattern projector (patent pending), biometric system and haul-truck tracking software; embedded software and project management.

SELECTED PROJECTS — videos at morg1207.github.io

- RB1 warehouse autonomy** — ROS 2 Humble, **tested on the real robot**: Cartographer mapping, localization and Nav2, with a BehaviorTree.CPP mission to find the charging station, detect a shelf, lift it and deliver it; Vue.js web control panel.
- AgroBot** (technical advisor, thesis) — weed detection and mapping in cacao plantations: **YOLO** on a **Jetson Nano**, outdoor navigation, SLAM and Kalman-filter sensor fusion.
- LeonardoBot** — mecanum robot with telescopic arm for interior wall painting, for construction startup **Smart Painter**. Autonomy & simulation lead: navigation, Gazebo Sim environment, mecanum control and painting routines.
- RRbot** (project lead) — scaled autonomous forklift, real prototype: pallet detection with Intel RealSense D435i, 2x LiDAR, Jetson Nano, SLAM and navigation.
- Berry Bot** — autonomous navigation robot on ROS; **1st place, International Startup Competition (ISC)**.
- ROS CI pipelines** — Jenkins + Docker pipelines that build and run automated tests for ROS 1 and ROS 2 waypoint-following packages.

EDUCATION, TEACHING & CERTIFICATIONS

B.Sc. in Mechatronics Engineering — National University of Engineering (UNI), Lima, Peru, 2022. **Languages:** Spanish (native) · English (intermediate).
Instructor, ROS Robot Programming and ROS + AI programs — Information Technology Office (OTI), UNI, 2025.
Certifications: Robotics Developer Masterclass (The Construct, 2024) · Robot Programming with ROS (The Construct) · Embedded C · ARM Microcontrollers.